15 Here are the costs of the same type of batteries in two shops.

Shop **A**	Shop **B**
Pack of 4 batteries £1.60	Pack of 6 batteries £2.70

Harry needs to buy at least 30 batteries.
He assumes that he has to buy batteries in whole packs.
Harry wants to buy the batteries as cheaply as possible from the same shop.

(a) Which shop should he buy the batteries from, shop **A** or shop **B**?
You must show all your working.

Shop A: 8 packs give 32 batteries; 8 × £1.60 = £12.80.

Shop B: 5 packs give 30 batteries; 5 × £2.70 = £13.50.

£12.80 < £13.50, so choose shop A.

4 marks total
Find whole-pack counts
A £12.80
B £13.50
Choose A

(4)

Harry's assumption is wrong.
He can buy single batteries for 40p each in shop **A** and for 45p each in shop **B**.

(b) Does this affect which of these two shops Harry should buy the batteries from?
Give a reason for your answer.

A: 7 packs + 2 singles = £11.20 + £0.80 = £12.00.

B: 5 packs = £13.50. Still shop A.

1 mark
Still shop A
Compare £12 with £13.50

(1)

(Total for Question 15 is 5 marks)

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